

BookMyShow: Monitoring, Observability & SLOs

Complete Architecture Specifications • Section F: Production Telemetry, Tooling Stack & Service Level Objectives

1. Core Key Performance Indicators (KPIs) to Monitor

KPI Name	Target Threshold	Operational & Business Relevance	Measurement Method
Booking Success Rate	> 99.5% completion	Monitors health of end-to-end checkout funnel from seat hold to ticket generation.	Prometheus: Confirmed / Initiated checkouts.
Average Booking Latency	p95 < 2,000ms (Avg < 800ms)	Measures responsiveness of payment authorization, ledger update, and seat commit.	OpenTelemetry: Checkout trace span duration.
Cache Hit Ratio	> 95% catalog / > 90% seats	Protects database primary and read replicas from ~50,000 read QPS flash surges.	Redis: keyspace_hits / (hits + misses).
Payment Failure Rate	< 1.0% system errors	Detects third-party payment gateway outages, timeouts, or webhook drops immediately.	Gateway logs: Failed payments / Total attempts.
Queue Backlog Size	< 1,000 msgs (lag < 3.0s)	Guarantees that async confirmation emails, SMS tickets, and reminders dispatch on time.	Kafka: Consumer group lag on booking topics.

2. Proposed Monitoring & Observability Tooling Stack

Prometheus (Metrics Engine) Scrapes microservices at 15s intervals for golden signals (latency, errors, QPS) and infra stats (Redis memory, DB pool saturation).	Grafana (Dashboards & Alerts) Real-time ops visualization for business health (tickets/sec, conversion funnel), SLO tracking, and PagerDuty alerting.	ELK Stack (Centralized Logging) Collects structured JSON logs with unified trace IDs across all services for rapid post-incident root cause analysis and financial auditing.
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3. Production Service Level Objectives (SLOs)

SLO 1: Booking Latency & Completion "99% of bookings complete in < 2 seconds." SLI: Valid checkouts completed with ticket in < 2,000ms. Target: 99.0% over rolling 30-day window (1% error budget). Escalation: Page on-call if budget burn > 2% in 1 hour.	SLO 2: High Availability & Zero Double-Booking "99.9% uptime with 0.00% double bookings." SLI: Catalog & seat map queries returning HTTP 200 in < 150ms. Target: 99.9% uptime (< 43.8 min monthly downtime). Invariant: Absolute 0.00% double-booking; fail-safe abort.
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4. Incident Severity & Escalation Matrix

Severity	Trigger Conditions	Response SLA	Automated Remediation Actions
P0 (Critical)	Booking success < 95%; double-booking detected; DB primary failover.	Immediate (< 5 min)	Route to backup gateway; activate read-only mode if consistency at risk.
P1 (Major)	Booking latency > 3s; cache hit < 85%; queue lag > 10,000 msgs.	< 15 minutes	Kubernetes HPA scales pods; throttle non-essential async consumers.
P2 (Minor)	Single notification provider degraded; isolated regional cache miss spike.	< 2 hours	Failover to secondary SMS/Email gateway; trigger cache warm-up worker.